

SOC Internship – Day 6

Elaborated Notes: SIEM (Security Information and Event Management)

1. What is SIEM?

SIEM (Security Information and Event Management) is a security platform that collects, correlates, stores, and analyzes logs from multiple sources in real time. Its main purpose is to help security teams identify suspicious activity and detect potential security threats.

In a SOC (Security Operations Center), SIEM acts as a central point where security events from different systems can be viewed and analyzed together. Instead of checking each server or security device separately, analysts can use the SIEM to examine events from multiple sources in one place.

2. Main Components of SIEM

- 1. Log Collection** – Collects logs and security events from different sources such as servers, firewalls, and endpoints.
- 2. Log Storage** – Stores collected logs so that analysts can review events and investigate activity.
- 3. Correlation Engine** – Compares and connects events from different sources to identify patterns that may indicate an attack.
- 4. Alerting System** – Generates alerts when defined rules or suspicious patterns are detected.
- 5. Dashboard** – Provides a visual interface where SOC analysts can monitor events, alerts, and security activity.

3. SIEM Workflow

The SIEM workflow describes how security information moves from different systems to the SOC analyst for investigation.

Step 1 – Collect Logs

SIEM gathers logs from multiple systems such as servers, firewalls, and endpoints.

Step 2 – Normalize and Store

Collected logs are organized into a common format and stored for analysis.

Step 3 – Apply Correlation Rules

The SIEM compares events and applies rules to identify relationships and patterns.

Step 4 – Detect Suspicious Activity

When activity matches suspicious patterns, the SIEM identifies a possible security event.

Step 5 – Generate Alerts

The alerting system notifies the SOC analyst so the event can be investigated.

SIEM Workflow in One Line

Log Collection → Normalization & Storage → Correlation → Detection → Alert

4. Real-World Detection Example: Brute-Force Attack

Suppose an attacker repeatedly tries to log in to an organization's servers using different passwords. Each failed login creates a log entry. A SIEM can collect these events from multiple servers and correlate them. If the number or pattern of failed login attempts matches a detection rule, the SIEM can generate a brute-force alert for the SOC analyst.

Example flow: Multiple failed logins → SIEM collects events → events are correlated → suspicious pattern is detected → brute-force alert is generated.

5. Popular SIEM Tools

Splunk – A popular SIEM platform used to collect, search, analyze, and monitor security-related data.

ELK Stack – A collection of Elasticsearch, Logstash, and Kibana used for collecting, processing, storing, searching, and visualizing logs.

IBM QRadar – A SIEM platform used by security teams to collect and analyze security events and investigate threats.

6. SOC Use Cases

Detect brute-force attacks – Identify repeated failed login attempts and suspicious authentication patterns.

Identify malware activity – Help analysts detect suspicious events associated with malware activity.

Monitor insider threats – Assist in identifying suspicious or unusual activity by users inside an organization.

Investigate incidents – Provide collected logs and related events that help analysts understand what happened during a security incident.

Assignment – Answers

1. Explain SIEM Workflow

Answer: SIEM workflow starts by collecting logs from multiple systems such as servers, firewalls, and endpoints. The collected logs are then normalized and stored. Next, the SIEM applies correlation rules to compare events and find relationships or patterns. If suspicious activity is detected, the SIEM generates an alert. The SOC analyst can then examine the alert and investigate the incident.

2. List 3 SIEM Tools

1. Splunk
2. ELK Stack (Elasticsearch, Logstash, Kibana)
3. IBM QRadar

3. Give One Real-World Detection Example

Answer: A brute-force attack can be detected using SIEM. For example, if multiple failed login attempts occur across servers, the SIEM collects these login events and correlates them. When the pattern indicates repeated unauthorized login attempts, it generates a brute-force attack alert for the SOC team to investigate.

Class Activity – SIEM Workflow Diagram

Multiple Systems

Servers + Firewalls + Endpoints



Log Collection



Normalize & Store Logs



Correlation & Analysis



Suspicious Activity Detected



Alert Generated



SOC Analyst Investigation

Key takeaway: SIEM helps a SOC centralize security logs, identify suspicious patterns, generate alerts, and support incident investigation.